CANDIDATE'S NAME: ……

SIGNATURE: ……

545/1

CHEMISTRY Paper 1

April 2026

2 hours

OXFORD MUSLIM HIGH SCHOOL KAWEMPE

END OF TERM ONE ASSESSMENT 2026

S.4

CHEMISTRY

Paper 1

2 hours

INNSTRUCTIONS TO CANDIDATES:

- This examination paper consists of two Sections; A and B.
- It has six examination items.
- Section A has two **compulsory** items. Responses to section A must be written in the spaces provided.
- Section B has two parts; I and II.
- Respond to only one item from each part.
- Responses to section B must be written in the answer sheet(s) provided.
- Respond to four items in all. Responses to each item in section B should start on a fresh page.
- Any additional item(s) answered will not be scored.

Use where necessary;

Ca = 40; C = 12; O = 16; 1 mole of a gas occupies 22.4 dm^3 at s.t.p.

SECTION A

Answer **both** *items from this section in the spaces provided*

Item 1

Two students of Oxford Muslim High School were picked at random by their chemistry teacher to look at the nature of their white shirt and determine which cleaning agents they use. Amos was putting on white shirt-stained brown and said he uses cleaning product x which is not the case with James who uses cleaning product **Y** as shown in figure 1 and figure 2 below. The students want to know more about the products and how it cleans and they have approached you for help.

Figure 1 cleaning product X

Figure 2 cleaning product Y

Task

a) Explain the problem Amos made in choosing cleaning product **X**

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b) How does the cleaning product Amos use works?

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c) What is the impact of using the cleaning products and how it can be mitigated?

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d) Evaluate the cleaning products used by Amos and James.

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ITEM 2

An industry wanted to produce lime for treatment of acidic soils. The production of lime involves heating limestone strongly which results into its decomposition according to the equation;

$CaCO_3$ (s) $\longrightarrow$ CaO (s) + CO_2 (g)

The industry is also interested to know how much gas is evolved during the process since the gas is useful. Before the production, an experiment was performed where 25g of limestone were heated until there was no further change. You have been contacted for help

Task:

As a learner of chemistry;

a) Explain the categories of the products.

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b) Suggest the properties of the gaseous product that make it useful in daily life

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c) Calculate the volume of the gaseous product measured at s.t.p. that was formed.

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d) Explain the impact of one of the products above on environment.

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SECTION B

Part I

Answer **one** *item from this part in the answer booklet(s) provided.*

Item3.

In Uganda, copper wires are used mainly for transmitting electric power. To ensure the availability of wires, the government is considering setting up a copper production plant in a certain area. However, the residents of the area need to be sensitized about the industrial processes, social benefits, side effects and how they can be overcome.

Your head teacher has been identified to sensitize the residents.

Task:

As a learner of Chemistry, prepare a write-up that your head teacher will use to sensitize the residents.

Item 4.

Cement is one of the most commonly used building materials. In order to meet the high demand of cement in Uganda, many cement factories have been set up across the country, one of which is shown in figure 2. You are part of the Chemistry class that visited one of the factories and the process of producing cement was explained to the class. You have been invited to a radio talk show to explain how cement is produced. After your presentation one of the callers wants to know whether there is any impact associated with the process you have talked about.

Fig. 2

Task: Make a write-up of the sensitization message up to the end of the talk show.

Part II

Answer **one** *item from this part in the answer booklets provided.*

Item 5.

Daniel, a cattle keeper, grazes his cattle on a community land. During the dry season, he practices bush burning and also takes his cattle to drink water from the community water source. Daniel's practices have raised concern in the community. The area chairperson has organized a meeting to create awareness for Daniel and the community.

Task:

As a learner of Chemistry make a write-up of the message the chairperson will present to the community.

Item 6.

During her School holidays, Huzair and Jane visited their aunt who works at a stone quarry. They noticed that explosives were being used to blast big rocks to form small stones (aggregates) and there was a lot of dust rising into the air as shown in figures **3** and **4**

Fig. 3 **Fig. 4**

Besides the social benefits, Huzair and Jane were concerned about what would happen to the site and community if the activity continued over time.

Task:

As a learner of Chemistry, make a write-up to respond to Huzair and Jane's concern.

END

"NO ONE WAS BORN KNOWING CHEMISTRY, IT'S ALL ABOUT GIVING IT TIME AND NOT GIVING UP"

Name……..…….

Signature……

(Success only comes to those who prepare for it, therefore do your very best)

545/1

CHEMISTRY

Paper 1

APRIL 2026

2 hours

OXFORD MUSLIM HIGH SCHOOL KAWEMPE

END OF TERM 1 ASSESSMENT 2026

2 Hours

S.3

INSTRUCTION TO LEARNERS:

- This examination paper consists of two Sections; A and B.
- It has six examination items.
- Section A has two **compulsory** items. Responses to section A must be written in the spaces provided.
- Section B has two parts; I and II.
- Respond to only one item from each part.
- Responses to section B must be written in the answer sheet(s) provided.
- Respond to four items in all. Responses to each item in section B should start on a fresh page.
- Any additional item(s) answered will not be scored.

SECTION A

Respond to all items in this section in the spaces provided

Item one

Two students of Oxford Muslim High School were picked at random by their chemistry teacher to look at the nature of their white shirt and determine which cleaning agents they use. Peter was putting on white shirt-stained brown and said he uses cleaning product x which is not the case with Titus who uses cleaning product **Y** as shown in figure 1and figure 2 below. The students want to know more about the products and

how it helps in cleaning and they have approached you for help.

Figure 1 cleaning product X

Figure 2 cleaning product Y

Task

a) Explain the problem peter made in choosing cleaning product **X**

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b) How does the cleaning product peter use works?

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c) What is the impact of using the cleaning products and how it can be mitigated?

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d) Help the students to know the effectiveness of the cleaning products.

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Item two.

A group of S3 learners were faced with a unique solid substance, **X**, which they suspected to be an element. The element could burn in air to form a solid product. One of them picked interest in what could be the chemical formula of the oxide of the element. However, he did not know how to determine the formula. When they contacted the laboratory technician, he gave them the atomic number and mass number of **X** as 12 and 24 respectively, and the symbolic representation of oxygen as $^{16}_{8}O$

Task

Help the learners to;

(a) understand the nature of substance **X**

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(b) determine the chemical formula of the oxide of **X**

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(c) Show how bonding occurs leading to formation of the oxide of X.

SECTION B
PART I

Respond to only **one** *item from this part in the answer sheets provided.*

Item 3.

Due to wide range of uses, ethanol is highly demanded country wide, making the government to clear an investor to set up ethanol manufacturing plant in one of the areas around Rukiga district due to the availability of raw materials like cassava, bananas and sorghum. However, the locals are against the proposal claiming that they don't know how the process of production will be carried out, the economic and social benefits of manufacturing plant and that it may have a negative impact on their land and the environment of which they may need to be informed on how it will be mitigated.

Task

Write details of the key issues you will present during the talk show.

Item 4

An investor is preparing to build an Oxygen producing company in Mbarara City as a means to develop the community. The chairperson of one of the areas where the plant will be established has requested that they hold a meeting with the scientists to work in the industry so as to know more how they will produce the oxygen gas and how the youths of the community will benefit from having the industry in their community. The elderly people also want to know how dangerous the process of production of oxygen will be to the environment.

Figure 2 industrial manufacture of oxygen gas

Task: Make a write up of message you will present in the meeting upon your invitation to the meeting as a scientist from Oxford Muslim High school science club addressing the concerns of the people in Mbarara city.

PART II

Respond to only **one** *item from this part in the answer sheets provided.*

Item 5.

Air is an important natural resource that should be conserved. There is improper waste management in some homes and schools involving open dumping and burning of garbage. These activities release toxic fumes into the atmosphere hence affecting the air quality. The head teacher has tasked the school Science Club to organize a sensitization campaign aimed at mitigating the challenge. The theme of the campaign is **AIR QUALITY IS EVERYONE'S RESPONSIBILITY**. You have been chosen to speak to the rest of the students regarding the theme. Figure 4 shows rubbish being burnt in a residential area.

Figure 4 Burning of rubbish

Task

Make a write up of the message you will present.

Item 6

A youth group has a washing bay along the river. As they wash vehicles, the waste water drains into the river. When senior one students from a nearby school went for a chemistry study tour to river, they found the youth busy with their routine work. One of the students noticed that the water body around the washing bay looked abnormal. The students expressed worry about the safety of the people around the river and drew attention of the area authority. In response, the local authority has called for a meeting with the youth. You have been selected to give a sensitization talk.

Task

Write details of the key issues you will address during the sensitization talk.

END

Enjoy this music of reasoning